Youwei Huang

Technology Director | Multimodal LLMs, Robotics & Web3

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PROFILE

I am the Technology Director at Shenzhen Yuewa Innovation Technology Co., Ltd. My research focuses on multimodal LLMs, robotics, and Web3. I continuously contribute to open-source projects, invention patents, and academic papers, bridging the gap between research and industry.

PROFESSIONAL EXPERIENCE

Shenzhen Yuewa Innovation Technology Co., Ltd. (LeapWatt) — Technology Director, Supervisory Board Member

Feb 2025–Present | Shenzhen, China; Hong Kong SAR

- **Contributions:** As an early technical leader, led the development of the **CoCoWa quadruped robot** and an **AI photography robotic arm**; secured the company's **angel round** from **top-tier investors** including **Meituan Strategic Investments, XVC, and Plum Ventures**.
- **Responsibilities:** Lead the company's AI Department; define core technical solutions and roadmaps; overcome deployment and runtime challenges for LLMs, VLMs, VLAs, and diverse CV models on NPU-based edge devices, including quantization, pruning, and operator-level optimization.

Institute of Intelligent Computing Technology, Suzhou, CAS — Lead Research Engineer

Suzhou Branch, Institute of Computing Technology, Chinese Academy of Sciences

May 2020–Dec 2024 | Suzhou, China

- **Responsibilities:** Led a research team advancing blockchain, big data, and AI; co-authored patent applications and academic publications; translated research outcomes into industrial applications.

Suzhou Zhongke Lelian Information Technology Co., Ltd. — Director of Blockchain Technology

IICT-incubated portfolio company

Jan 2022–Oct 2022 | Suzhou, China

- **Responsibilities:** Led blockchain technology R&D and commercial project development.

Shanghai LeMiao Network Technology Company — Software Engineer

Jun 2018–Aug 2019 | Shanghai, China

- **Responsibilities:** Developed web applications and online games.

EDUCATION

- Macau University of Science and Technology — Ph.D. Candidate, Jan 2022–Jan 2023 (withdrew). Research: deep learning, blockchain, and software engineering.
- Monmouth University — M.S. in Software Engineering, Sep 2019–Aug 2020. ABET-accredited program.
- Changshu Institute of Technology — B.Eng. in Software Engineering, Sep 2013–Jun 2018; one-year entrepreneurship gap.

SELECTED RESEARCH & PROJECTS

CoCo / GhostShell — Concurrent Embodied Programming

Feb 2025–Present | Shenzhen Yuewa Innovation Technology Co., Ltd.

- Designed and developed CoCo, a spider-like desktop quadruped for natural-language interaction and on-the-fly actions.
- Proposed GhostShell, an LLM-driven streaming and concurrent embodied programming approach using streaming function calls, multi-channel scheduling, and dynamic prompt trees.
- Paper: “GhostShell: Streaming LLM Function Calls for Concurrent Embodied Programming” (2025).
- Demo: coco-robot.github.io/GhostShell | Video: Bilibili

LeChat / UniAI — Document Parsing and Conversational Platform

Mar 2023–Dec 2024 | Institute of Intelligent Computing Technology, Suzhou, CAS

- Built an LLM-driven platform that parses Office documents, recognizes images, interprets semantics, and generates text, images, and charts.
- Launched UniAI, integrating multiple LLMs with knowledge graphs; transferred the foundation into three commercial projects.
- Received the “AIAC 2024 Outstanding Individual in Artificial Intelligence” award and secured 3 invention patents.

AI for Web3 Software Engineering — Smart Contract Security

Jan 2022–Dec 2025 | Macau University of Science and Technology

- Researched detection, repair, and generation of smart contracts using deep learning.
- Developed SmartIntentNN for unsafe development-intent detection and SmartBERT for smart-contract representation learning.
- Resources: Hugging Face | GitHub | Website

LeChain — Blockchain for Medical and Healthcare Applications

May 2021–Oct 2023 | Institute of Intelligent Computing Technology, Suzhou, CAS

- Developed a consortium blockchain enabling trusted medical-data exchange among hospitals, patients, insurers, and authorities.
- Named a “2022 Jiangsu Province Blockchain Industry Development Pilot Demonstration Project”; published technical white papers, secured 5 patents, and incubated Zhongke Lelian.

Precise Epidemic Prevention and Control

May 2020–May 2021 | Institute of Intelligent Computing Technology, Suzhou, CAS

- Developed GeoHash-based regional safety metrics and Generic Wireless Base technology for anonymous contact-network reconstruction using Bluetooth, Wi-Fi, and cellular signals.
- Supported government COVID-19 response; published at IEEE DSA 2022, received the 2021 Suzhou Youth Talent Support selection, and secured 4 patents.

SELECTED PUBLICATIONS

1. Y. Huang, J. Li, B. Hu, et al. “Detecting Malicious Intents in Smart Contracts with Pre-trained Programming Language Models.” ACM PROMISE 2026, pp. 41-50.
2. J. Gong, Y. Huang, B. Yuan, et al. “GhostShell: Streaming LLM Function Calls for Concurrent Embodied Programming.” arXiv:2508.05298, 2025.
3. B. Hu, Y. Huang, and X. Sang. “A Chain Division-based Vertical Scalability Method of Hyperledger Fabric.” Chinese High Technology Letters 35(9), 2025.
4. Y. Huang, S. Fang, J. Li, et al. “Deep Smart Contract Intent Detection.” IEEE SANER 2025.
5. Y. Huang, S. Fang, J. Li, et al. “SmartIntentNN: Towards Smart Contract Intent Detection.” arXiv:2211.13670, 2022.
6. Y. Huang, F. Lu, X. Sang, et al. “Precise Epidemic Control based on GeoHash.” IEEE DSA, pp. 1040–1048, 2022.
7. Y. Huang and Y. Gao. “Financial Data Forecasting System Based on Recurrent Neural Network.” Software Guide 18(1), 2019.
8. J. Zhou, G. Zhao, and Y. Huang. “Hidden Markov Model-based Multi-node Fusion Decision Concurrent Troubleshooting.” Application Research of Computers 32(8), 2015.

PATENTS

Co-inventor on 12 published invention patents spanning knowledge-graph retrieval augmentation, enterprise-industry-chain association, technology supply-demand matching, smart-contract representation and intent detection, blockchain medical insurance and evidence preservation, NFT certificates, and wireless-signal-based epidemic monitoring/contact tracing. Patent portfolio: CN119808914A, CN119204179A, CN118780853A, CN118821849A, CN119205357A, CN117709505A, CN119210677A, CN116662991A, CN114201735A, CN113163336B, CN112203229B, and CN112037924B.

SELECTED HONORS & CREDENTIALS

- AIAC AI Excellence Award — 2024 Outstanding Individual in Artificial Intelligence, Sep 2024.
- Jiangsu Province Intermediate Professional Technical Qualification — Engineer, Aug 2024.
- 2022 Jiangsu Province Blockchain Industry Development Pilot Demonstration Project, Aug 2022.
- 2021 Youth Talent Support Project, Suzhou Association for Science and Technology, Dec 2021.
- Monmouth University TechPitch Competition — Second Place, Oct 2019.
- Second Prize for Outstanding Undergraduate Graduation Project, Jun 2018.
- China College Student Computer Design Competition — Third Prize, Aug 2015.

SELECTED MEDIA COVERAGE

- “Shenzhen: Building a Pioneer City for Artificial Intelligence”, China Reform Daily, Jun 2025.
- “Blockchain Smart Contract Hidden Risk Research Makes Latest Progress”, China Enterprise News, Apr 2025.
- “Huang Youwei: From Code to Honor, A Full-Stack Engineer's AI Innovation Journey”, Sohu Business, Nov 2024.
- “Multi-model Fusion and Knowledge Graph, Driving the New Wave of Large Model Industrialization”, China.com, Oct 2024.

TECHNICAL SKILLS

AI/ML: PyTorch, ONNX, TensorFlow, Hugging Face, OpenCV, CUDA, Multimodal Learning, Computer Vision, Edge Intelligence

Software & Robotics: Linux, Vim, Python, TypeScript, JavaScript, ROS 2

Web3: Solidity, smart-contract analysis, Hyperledger Fabric, consortium blockchain